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Accelerating technology companies

SmartCoat

Process mining explained by an example

The logistics process at SmartCoat Inc.

Episode 2 *(out of 8)*: **process discovery**

What’s on this week?

Process discovery

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What preceded…

What preceded…

Marie, CEO of SmartCoat Inc., asked us to analyze and make recommendations for the logistics process via **process mining** and **data analytics** techniques ... Just by looking at the data in SmartCoat's ERP system!

In the previous episode, Marie shortly portrayed her company and gave more information about the logistics process at SmartCoat Inc.

Have you **missed** the **first episode**? Click on Marie… and you will be redirected to the first episode!

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Cédric, your horsum guide

Cédric, your horsum guide

Cédric
Consultant

Hi, I am Cédric. I am a consultant at horsum!

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I will guide you through the next episodes of the SmartCoat case.

In this episode, I will explain you more about the **event log** I have created out of SmartCoat's ERP data. I will also show you the **real logistics process** I have discovered via process mining.

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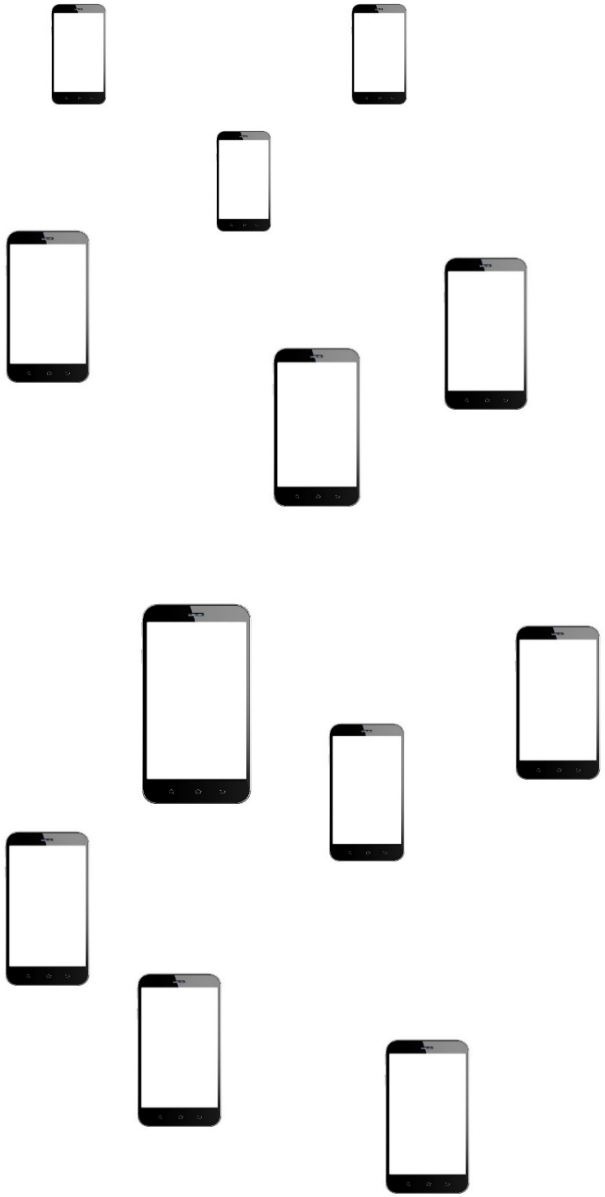
Event log

Event log

In the ERP system, I selected all **uncoated smartphones** that were **received** during the first half of March 2016.

35 smartphones have been received in that period. They all have a unique serial number: phone 3651, phone 3652, … up until phone 3685.



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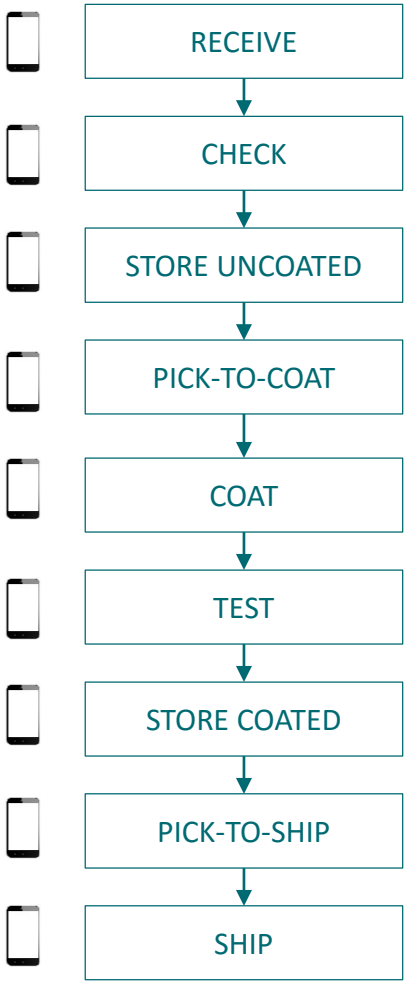
Event log

I would like to follow the journey of these smartphones throughout the logistics process. I would like to know **what exactly happened** with these smartphones!



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Event log

I managed to compile an event log out of the data stored in SmartCoat's ERP system.

You can find my **event log** in **Excel** over here:

Event log
SmartCoat Inc.

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Event log

Let's have a look at the event log in detail. What happened with **smartphone 3651** for example? Below we see a part of the event log: the journey of Phone 3651 throughout the logistics process. We identify **what** type of **activities** were performed, **when** the activities were performed, **by whom** and **where**.

Case ID	Activity description	Timestamp	Resource	Location
Phone 3651	RECEIVE UNCOATED	2/03/2016 8:10	Arthur	INBOUND AREA
Phone 3651	CHECK OPERATION	2/03/2016 9:13	Arthur	INBOUND AREA
Phone 3651	CHECK DAMAGE	2/03/2016 9:25	Arthur	INBOUND AREA
Phone 3651	STORE UNCOATED	2/03/2016 10:08	Arthur	WAREHOUSE UNCOATED
Phone 3651	PICK-TO-COAT	15/03/2016 10:51	Jerome	WAREHOUSE UNCOATED
Phone 3651	START COATING	16/03/2016 15:14	Alix	COATING ROOM
Phone 3651	STOP COATING	16/03/2016 15:34	Alix	COATING ROOM
Phone 3651	TEST 1	17/03/2016 16:34	Edward	TESTING ROOM
Phone 3651	TEST 2	18/03/2016 10:34	Edward	TESTING ROOM
Phone 3651	TEST 3	18/03/2016 14:34	Edward	TESTING ROOM
Phone 3651	STORE COATED	18/03/2016 16:04	Jerome	WAREHOUSE COATED
Phone 3651	PICK-TO-SHIP	24/03/2016 9:33	Jerome	WAREHOUSE COATED
Phone 3651	SHIP COATED	24/03/2016 15:33	Jerome	OUTBOUND AREA

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Event log

I also identified the following information in the ERP system:

- The related smartphone **brand**;
- The related **retailer**;
- The **function** of the resource;
- The **cost** of the **activity**.

For richer analysis purposes, I have included this information in the event log too.

Brand
MePhone
Simsong

Customer (retailer)
Wallsmart
Callhouse
Phonemarket

Alix
Coater

Edward
Test engineer

Elise
Test engineer

Tom
Warehouse manager

Jerome
Warehouse assistant

Arthur
Warehouse assistant

Activity description	Estimated cost price
RECEIVE UNCOATED	€ 0,50 per activity
CHECK OPERATION	€ 2,00 per activity
CHECK DAMAGE	€ 0,50 per activity
RETURN	€ 0,50 per activity
STORE UNCOATED	€ 0,10 per phone per 24 h
PICK-TO-COAT	€ 0,10 per activity
COATING	€ 20,00 per hour
TEST 1	€ 0,75 per activity
TEST 2	€ 0,25 per activity
TEST 3	€ 2,50 per activity
PROPOSE SCRAPPING	€ 0,05 per activity
EVALUATE SCRAPPING	€ 0,05 per activity
STORE COATED	€ 0,10 per phone per 24 h
PICK-TO-SHIP	€ 0,10 per activity
SHIP COATED	€ 0,50 per activity

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Process discovery

Process discovery

It is now time to discover the **real journeys** of the **smartphones** throughout the **logistics process**.

I **upload** the event log in the **process mining software** Disco.

Disco
by Fluxicon

Thanks to Disco, we are able to **immediately visualize** the real journeys of the 35 smartphones.

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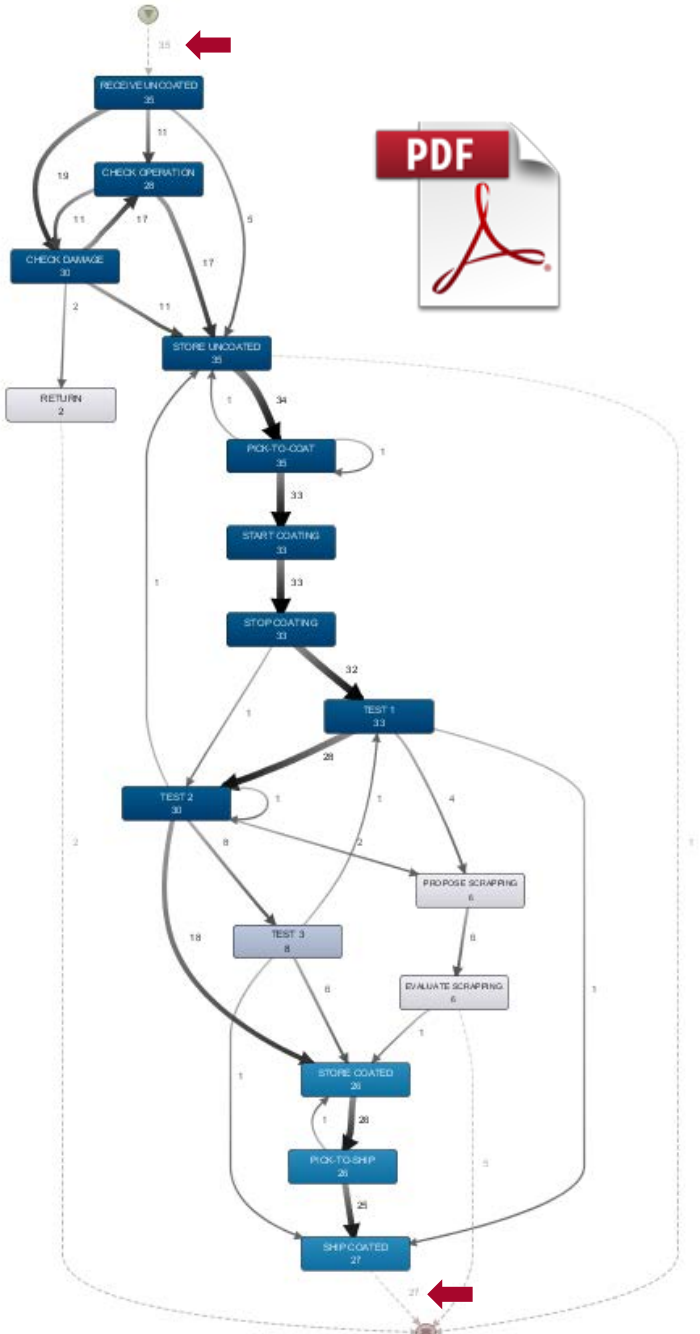
Process discovery

The visual of the real process map is shown on the right. **Is this visual too small for you?** Then click on the PDF icon to zoom in on the visual.

We immediately notice that the logistics process **starts** with **35 smartphones** and that **only 27 smartphones** are **finally shipped** to the retailer. What about the other 8 smartphones?

We also instantly see that the real process is not that linear, which means that the journeys are not always similar for all smartphones… Let's further investigate the event log!

Real process map - 35 smart phones



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Process discovery

Event log analysis	
Number of cases	35
Number of activities	16
Number of events	393
Start	02.03.2016 08:10:00
End	31.03.2016 15:33:57
Median case duration	17 d
Mean case duration	16,5 d
Number of variants	18

There are **35 cases**:

- Every case represents a unique smartphone;
- **16 activities** are discovered in the logistics process. These activities are shown as blue boxes in the process map.
- There are **393 events** in total. They represent the number of activities in total that are executed by the 35 smartphones.

The first activity was performed on March 2nd, the last one on March 31st. On average, the journey of a smartphone took **17 days**.

There are **18 variants**. This means that amongst all 35 smartphone journeys, 18 different paths were followed. I wonder whether one path occurs more frequent than the others. Let's investigate…

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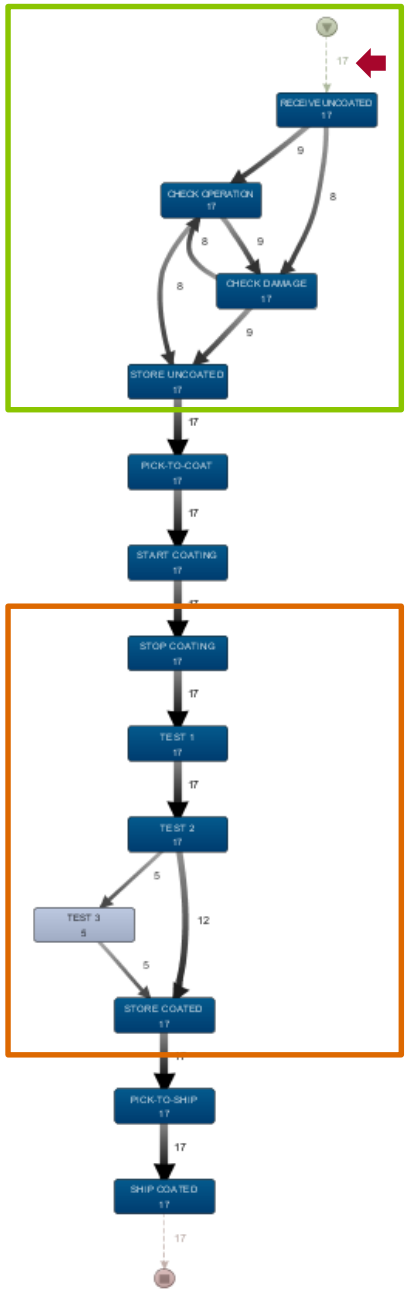
Real process map - top 3 variants

Process discovery

Top 3 variants	Cases	Events	Median duration	Mean duration
Variant 1	8	12	16 days, 18 hours	16 days, 22 hours
Variant 2	5	13	19 days, 5 hours	18 days, 20 hours
Variant 3	4	12	14 days, 17 hours	15 days, 4 hours
Total	**17**			

Above we see that the **3 most frequent paths** (variants) are followed by 17 smartphones. Put differently, these 3 paths cover 48,6% of all 35 smartphone journeys.

On the right, we see a visual of the process map of these 3 most frequent paths. I'm zooming in on two parts of the process on the next slides.



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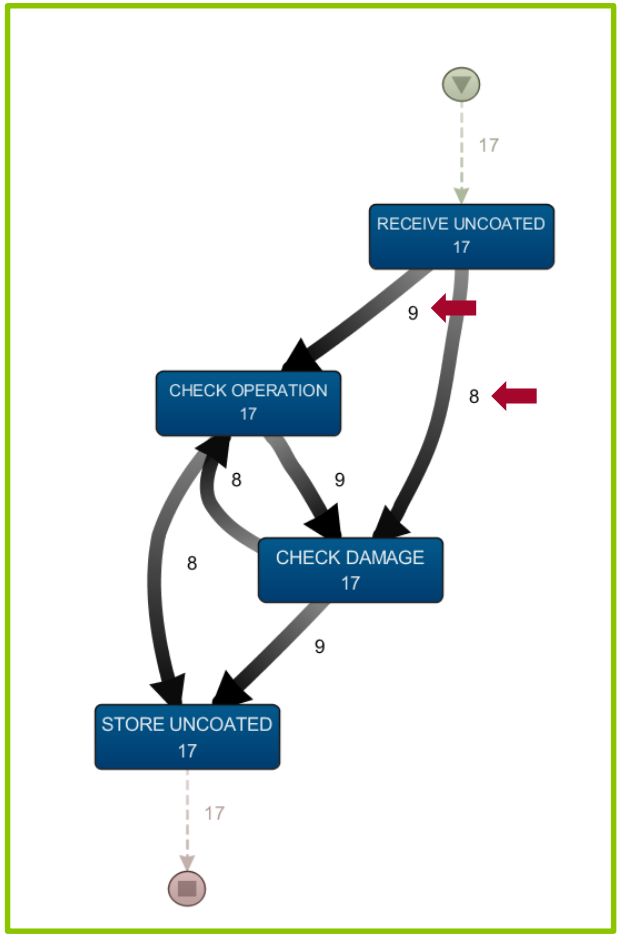
Process discovery



In the first episode, Marie told us that all received **uncoated** smartphones need to be **checked** before storage.

On the left, we see that the 17 smartphones (following the most frequents paths) are checked. There are apparently two types of checks that need to be done: “check operation” and “check damage”. We see that for 9 smartphones the “check operation” activity is done prior to the “check damage” activity. The other 8 smartphones are checked in reverse order. I wonder whether there are guidelines on the **sequence** of the checks.

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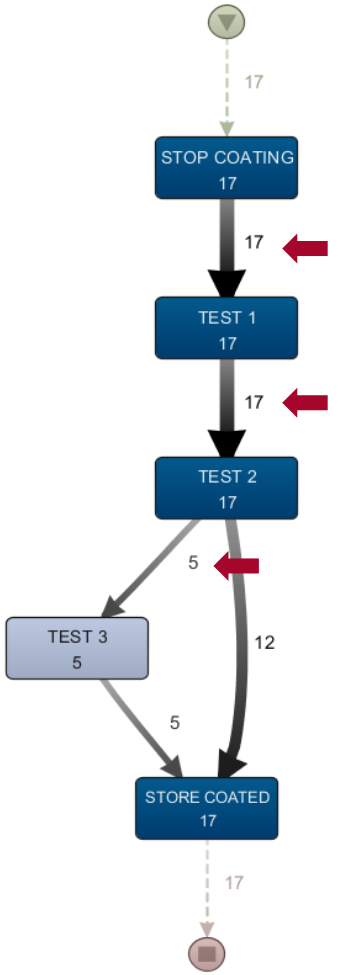
Process discovery

Coated smartphones need to be **tested** before storage. The process map on the right shows that “test 1” and “test 2” are done for all 17 smartphones. “Test 3” is only performed for 5 smartphones.

Marie told us that customer Wallsmart requires an additional test. It is very probable that the 5 smartphones are Wallsmart phones.

Let’s check… and indeed… they concern Phone 3651, 3562, 3600, 3669 and 3675… they are all Wallsmart phones!



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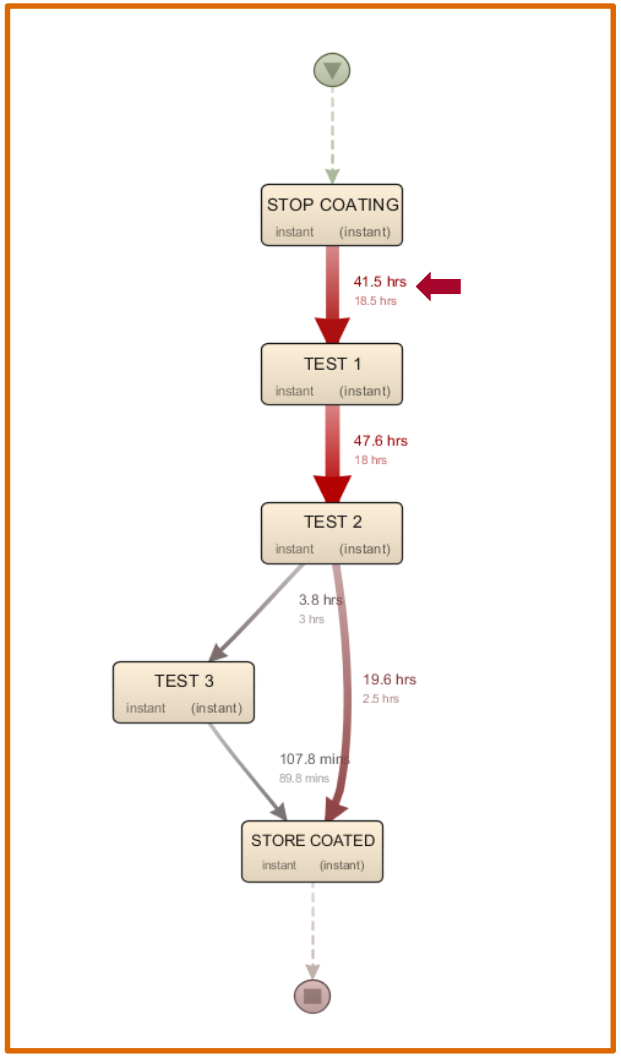
Process discovery

The process mining software Disco also allows us to visualize the performance. For instance, we can check the mean time between two activities or the minimum time a certain case took.

Marie told us that, once a smartphone is coated, a *cooldown* period of *24 hours* is required before the tests can start. On the right, we can see that the average time between “stop coating” and “test 1” for the 17 phones takes on average 41,5 hours. For one case, it only took 18,5 hours (minimum time). So, the cooldown period was not respected for that case.

Let’s further analyze this in episode 3: process deviations.
Do you come along?



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Do you also want to know **how** and **what process deviations** Cédric will identify via **process mining** and **data analytics** techniques?

Watch the episode next week!

Planning

April 7th, 2016	Episode 1: **introduction**
April 14th, 2016	Episode 2: **process discovery**
April 21st, 2016	Episode 3: **process deviations**
April 28th, 2016	Episode 4: **benchmarking**
May 5th, 2016	Episode 5: **bottlenecks**
May 12th, 2016	Episode 6: **interactions**
May 19th, 2016	Episode 7: **process costs**
May 26th, 2016	Episode 8: **prediction** and **real-time**

Or check our website! www.horsum.be

Questions? Contact us!

Contact us!

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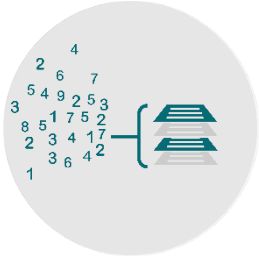

Data analytics

Process mining

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